

PGR Intelligence - Frontend Experience

Detailed interaction and UI specification for integrating the upgraded AI capabilities into the current Next.js 14 + Tailwind + shadcn/ui Redwood Professional frontend.

Frontend information architecture, wireframes and states	PGR Intelligence Programme
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Built around the existing PGR platform: lifecycle, supervision, progression, funding, thesis, documents, workflows, Pattern Lab and Ask PGR.

1. Frontend goal - intelligence in the work, not beside it

The current application already has a clear left navigation, student marquee record, domain pages, Pattern Lab and a globally available Ask PGR launcher. Preserve that information architecture. The AI upgrade should appear as contextual intelligence strips, drawers, timelines and review cards inside the screens users already trust.

Current frontend anchor	Upgrade rule
Next.js 14 + Tailwind + shadcn/ui Redwood Professional	Implement intelligence as reusable components in the same design system, not a separate visual language.
Student record has collapsible domain sections	Place PGR Intelligence immediately below the student header; deep evidence stays in drawers/tabs.
Ask PGR is globally available	Keep Ctrl/Cmd+K and floating launcher; add student/cohort "case mode" when context exists.
Pattern Lab is an advanced governed workspace	Keep governance controls; improve student-facing visualisation without making the model look authoritative.
Row scoping is already core	Frontend must never imply existence of out-of-scope students/documents in autocomplete, evidence, similarity or alerts.

Consistent interaction language: Explain -> Evidence -> Simulate -> Prepare action. Users should learn these four verbs once and see them everywhere.

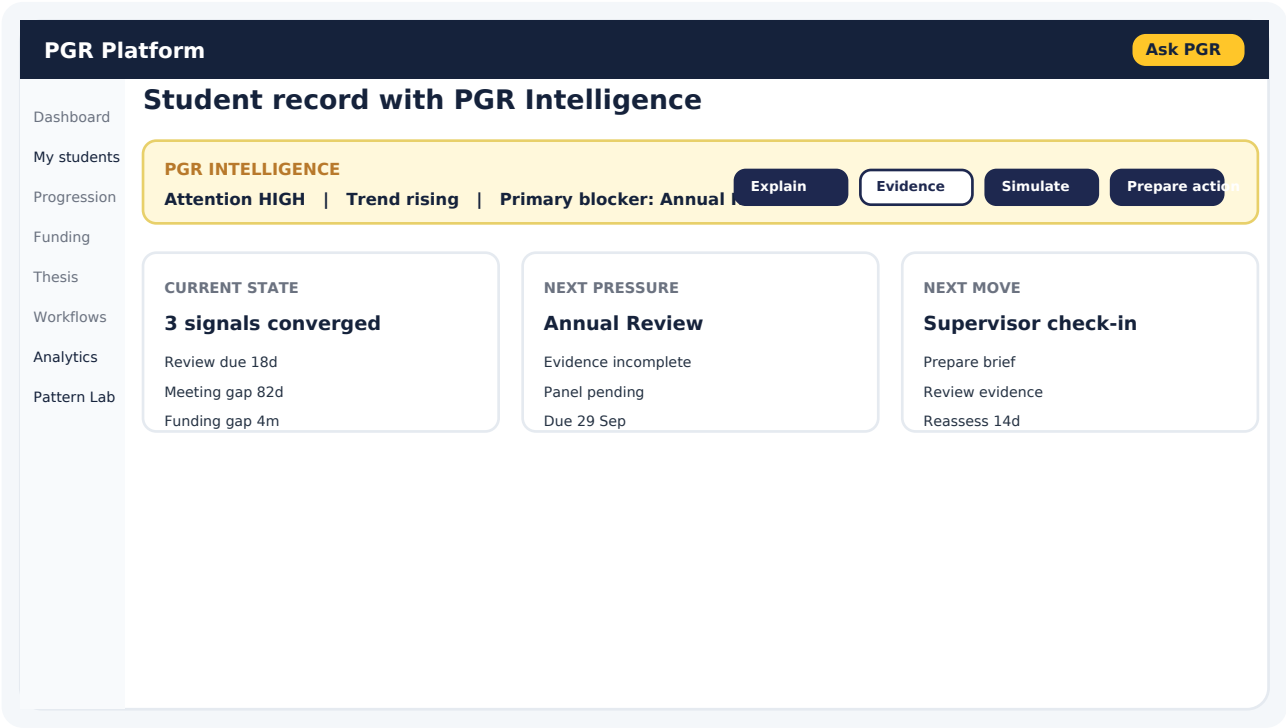
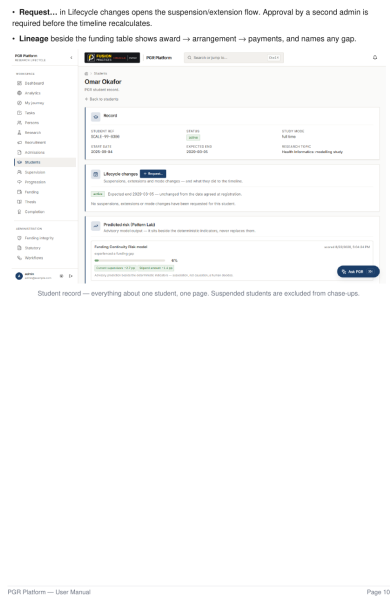
2. Visual language - align to the current platform

Use the existing navy/white professional shell and gold accent visible in the current PGR UI. AI should be visually identifiable but not neon or "magical". Status colours must retain their existing operational meaning.

Token / component	Meaning	Frontend treatment
Navy	Authoritative product UI	Headers, navigation, primary actions, text hierarchy
Gold	PGR Intelligence accent	Small intelligence label, active insight indicator, not every AI element
Green	Confirmed / completed / healthy	Verified evidence, completed intervention, healthy model state
Orange	Needs review / assumption / missing information	Document findings, model limitations, scenario assumptions
Red	Deterministic issue / deadline / failed check	Funding integrity error, overdue rule, explicit hard issue
Purple	Model / analytical signal	Pattern Lab probability, trajectory and sensitivity
Evidence badge	Source-linked fact	Small "verified" / "review" badge with source drawer

3. Student record - current anchor -> intelligence hub

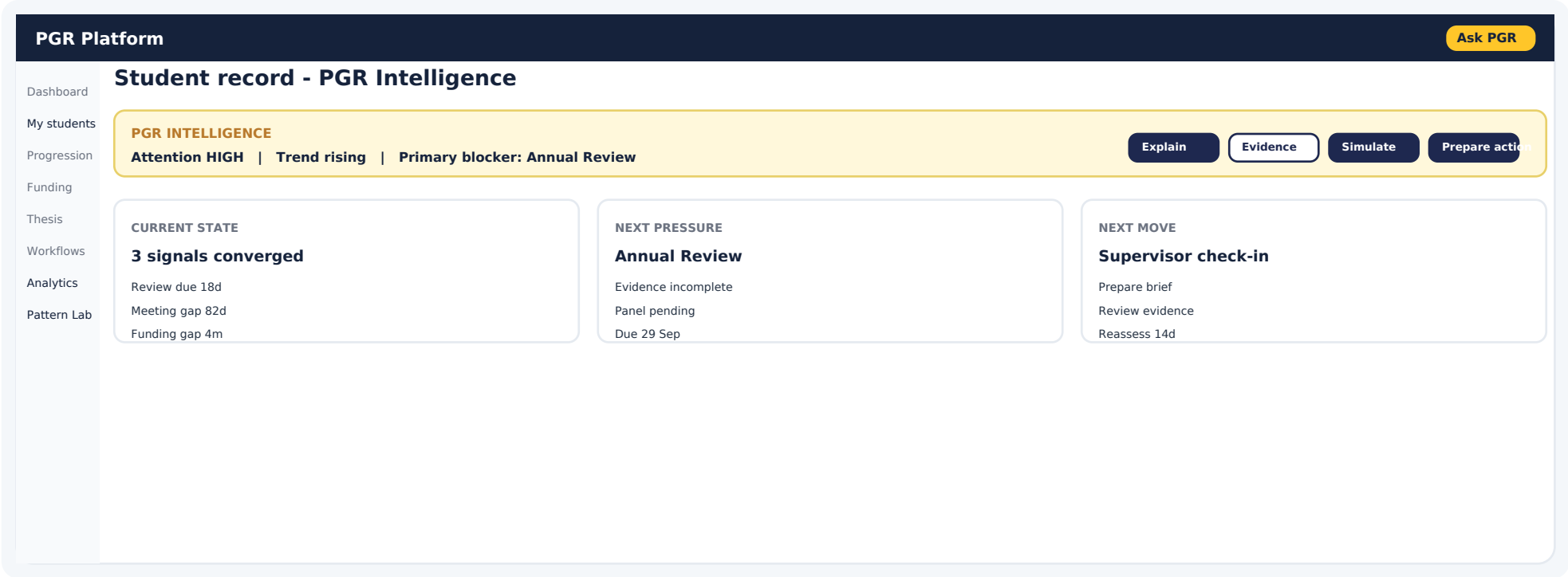
The student record is already the richest screen in the platform. Do not create a separate "AI student profile". Add one intelligence strip directly beneath the existing student header so the AI feels native to the record.



Current screen anchor	Upgrade direction
Current UI: Student record / lifecycle section from the supplied manual.	Target: keep the same navigation, route context and domain cards. Add intelligence in-line rather than creating a separate AI application.

4. PGR Intelligence strip - exact behaviour

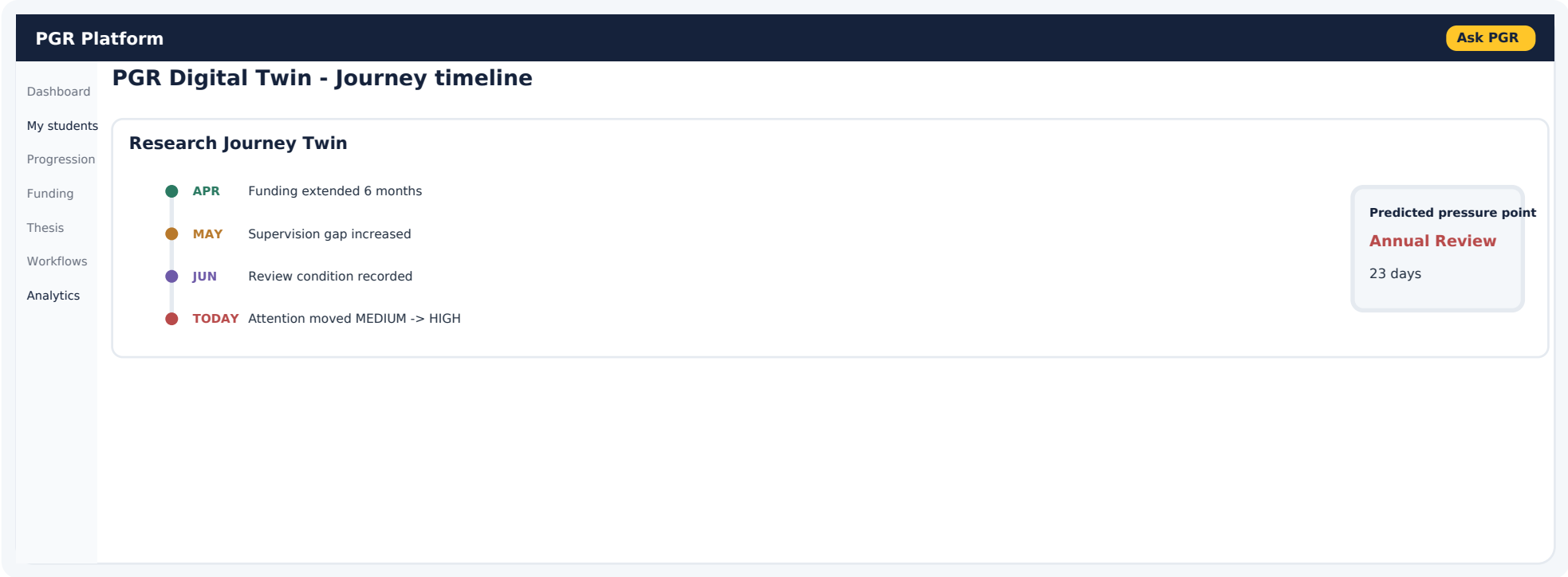
The strip is the compact entry point to the whole intelligence layer. It should answer what matters now in one line, then expose four actions. It must render deterministically before any narrative call completes.



Element	Behaviour
Attention	Derived priority state; clickable to show factors and source evidence.
Trend	Shows change from previous intelligence snapshot, not an unsupported emotional judgement.
Primary blocker	One best-supported blocker; if ambiguous, show "Multiple / needs review".
Explain	Concise narrative of why the state changed, linked to EvidenceClaims.
Evidence	Opens right-side Evidence Drawer with source rows/documents/model info.
Simulate	Opens scenario drawer only for supported, side-effect-free scenarios.
Prepare action	Opens intervention plan; never executes immediately.

5. Digital Twin timeline

The Twin should be a timeline/read model inside the student context, not a futuristic 3D avatar. The UI value is longitudinal change and cross-domain consequence.



Timeline event	Visual rule
Deterministic lifecycle change	Navy/green marker with source record and effective date
Model score change	Purple marker showing old -> new plus model-version boundary if relevant
Funding integrity change	Red/orange marker only when a deterministic check changes state
Supervision engagement change	Neutral contact marker + reason code
Intervention	Action marker with owner and completion status
Missing source	Hollow/orange marker explaining which conclusion is limited

6. Scenario drawer - compare current vs proposed

Scenario exploration should open from the student record/Twin in a right-side drawer or full-width modal. The user must be able to see exactly which values are deterministic and which are model sensitivity.

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Scenario preview - 12 week suspension

Scenario preview - no state changes, no notifications, no outbox events.

Field	Current	Scenario	Impact
Expected end	30 Sep 2027	23 Dec 2027	+12 weeks
Annual review	29 Sep 2026	22 Dec 2026	moves
Funding end	31 Jan 2027	31 Jan 2027	UNCHANGED

Model sensitivity is not a causal prediction. Funding is not assumed to extend.

UI requirement	Detail
Persistent banner	"Preview only - nothing has changed" at the top of the drawer.
Two result groups	Rule impact and Model sensitivity in separate cards.
Assumptions	Editable scenario inputs and explicit unchanged assumptions, especially funding.
Unavailable information	Show missing Finance confirmation / policy uncertainty rather than guessing.
CTA	"Start request from scenario" copies inputs to the real flow and revalidates them.

7. Weekly Review -> Intervention Queue

Upgrade the current review experience into an operations queue. Users should see why the student is prioritised and whether the next action belongs to the student, supervisor or institution.

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Weekly Review - Intervention Queue

This week: 12 students need review. 4 are driven by institutional blockers.

Priority	Student	Why now	Intervention type	Owner	Action
1	Marcus Bell	Review 18d + meeting 82d	Supervisor check-in	A. Ford	Prepare
2	Amina Shah	Funding ends before completion	Pending review	PGR Admin	Prepare
3	Noah Kim	Panel waiting 16d	Institutional unblock	Registry	Open

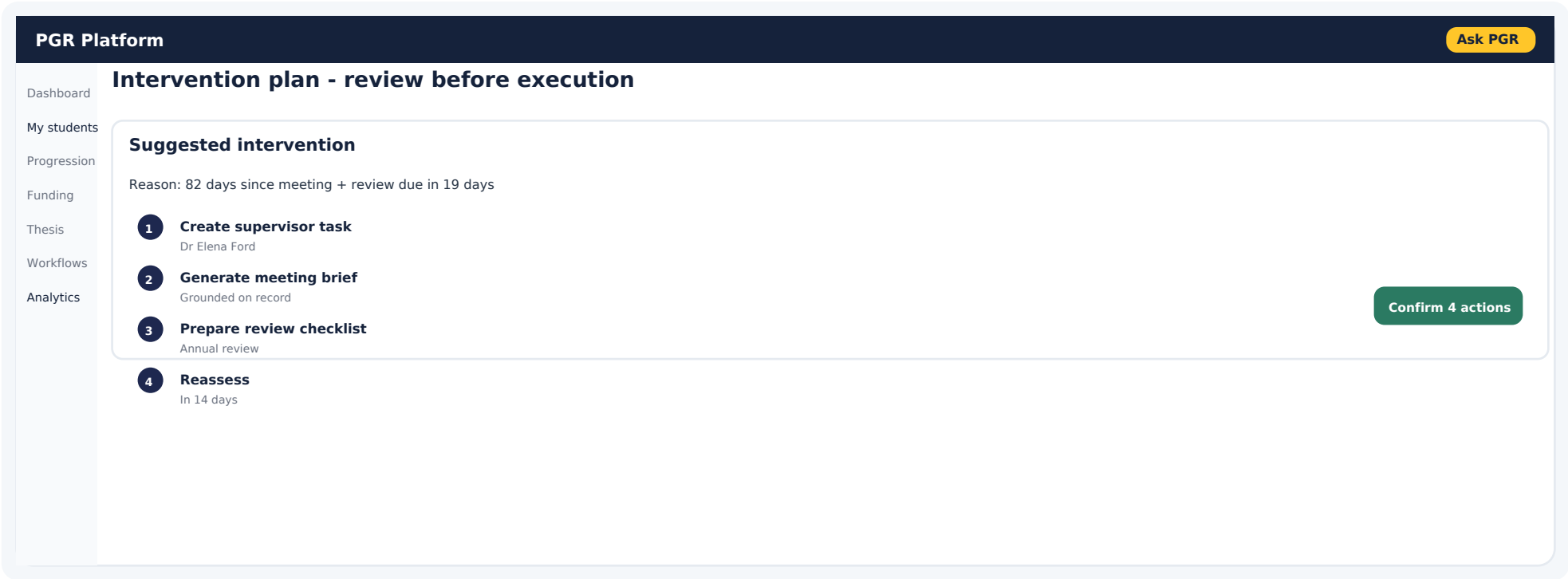
Systemic pattern detected

4 cases are waiting on panel confirmation rather than student action.

Add bulk filtering by intervention type, owner, deadline window and institutional blocker. Keep ranking explanation on row expand; do not force users into chat to understand the queue.

8. Intervention plan confirmation

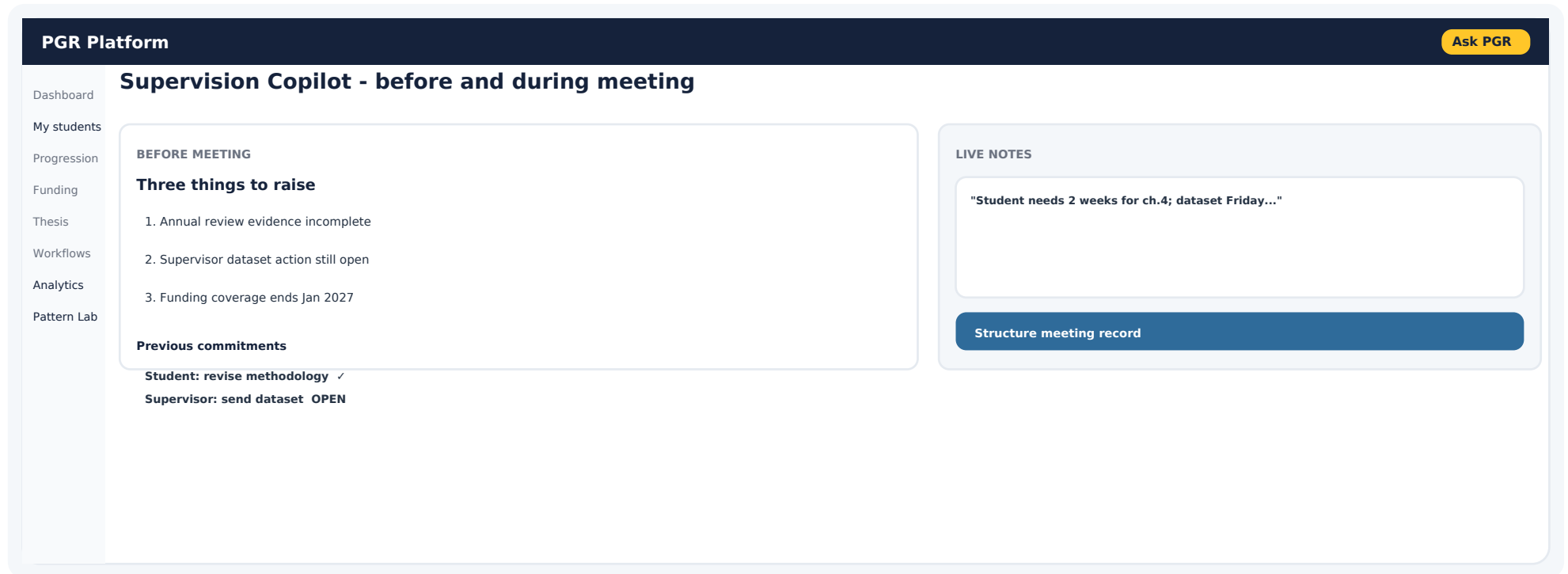
When a user selects Prepare, open a drawer containing the full proposed action bundle. This is the critical trust boundary between AI recommendation and workflow execution.



Action row	Required UI
Type	Human-readable action + target record
Owner	Resolved owner with edit control when allowed
Due date	Source label: policy / existing deadline / user-selected / not specified
Evidence	One-click source drawer
Dependency	Shows if another action must happen first
Execute checkbox	User can deselect individual action from the bundle
Confirmation footer	Exact count of actions + warning that current permissions/state will be rechecked

9. Meeting Brief -> Supervision Copilot (before + during)

Keep the existing one-tap Meeting Brief, but add previous commitments and a lightweight live-notes area. The supervisor should never have to copy the meeting context into Ask PGR manually.



The pre-meeting view should prioritise a maximum of three discussion points. Everything else sits under "More context". The live-notes box is optional and should not auto-save AI-extracted actions.

10. Supervision Copilot (after meeting)

After notes are submitted, show a structured review step. This is where AI turns shorthand into proposed commitments while preserving human ownership of the record.

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Review meeting record and commitments

Review extracted commitments before they become part of the record.

Decision

Chapter 4 target extended by 2 weeks

Student action

Update analysis chapter

Depends on dataset

Supervisor action

Send revised dataset

Owner: Dr Ford

Deadline

Not specified

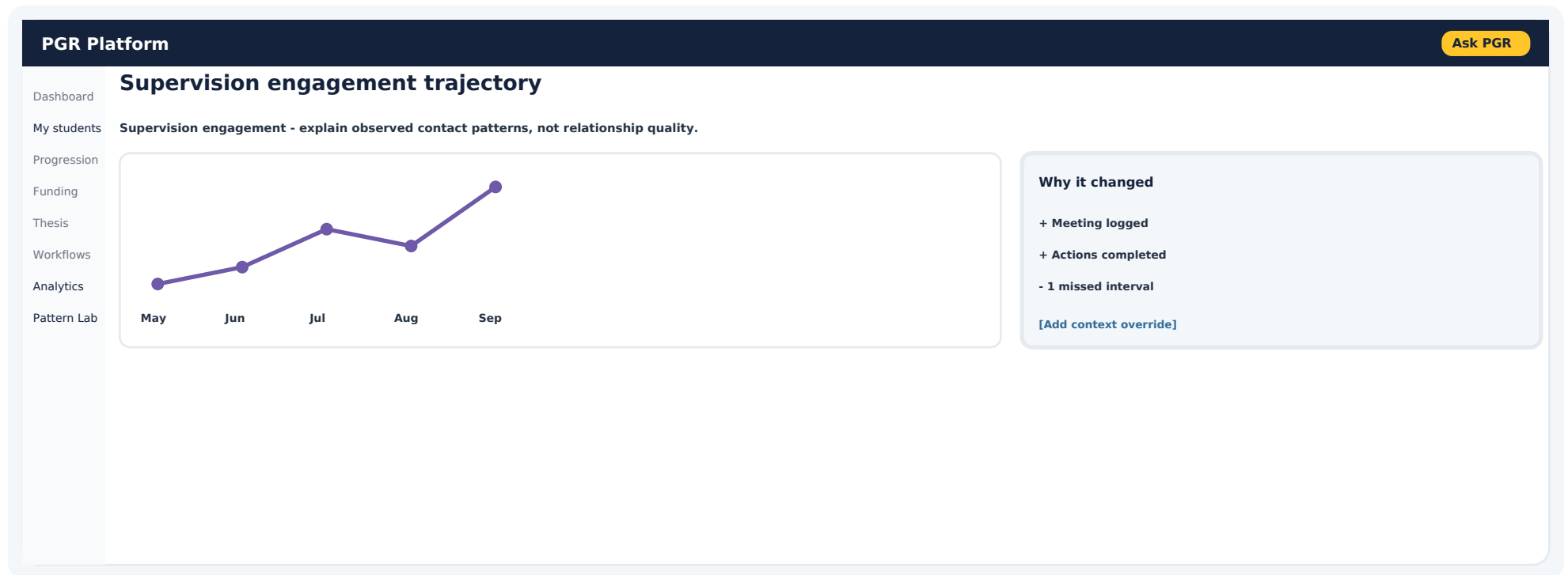
Do not invent

Confirm record

Use orange styling for inferred/proposed fields, green only after confirmation. Null dates should render as "Not specified" - never silently inferred from context.

11. Engagement Trajectory

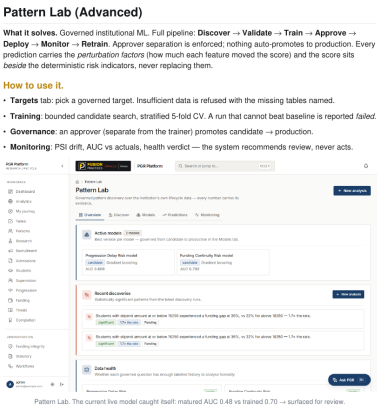
Replace a purely static relationship badge with a deeper trend view while keeping the quick badge on list/card surfaces. Call it Supervision engagement or Contact health, not relationship quality.



A small "Add context" action lets an authorised user record fieldwork, approved leave or missing historical meetings. Context becomes an auditable event and explains future signal changes.

12. Pattern Lab - current anchor -> risk storyline

Pattern Lab already has mature governance. The frontend upgrade should preserve that advanced workspace and change the student-level presentation from a single score into a trajectory with model health, driver change and safe sensitivity exploration.



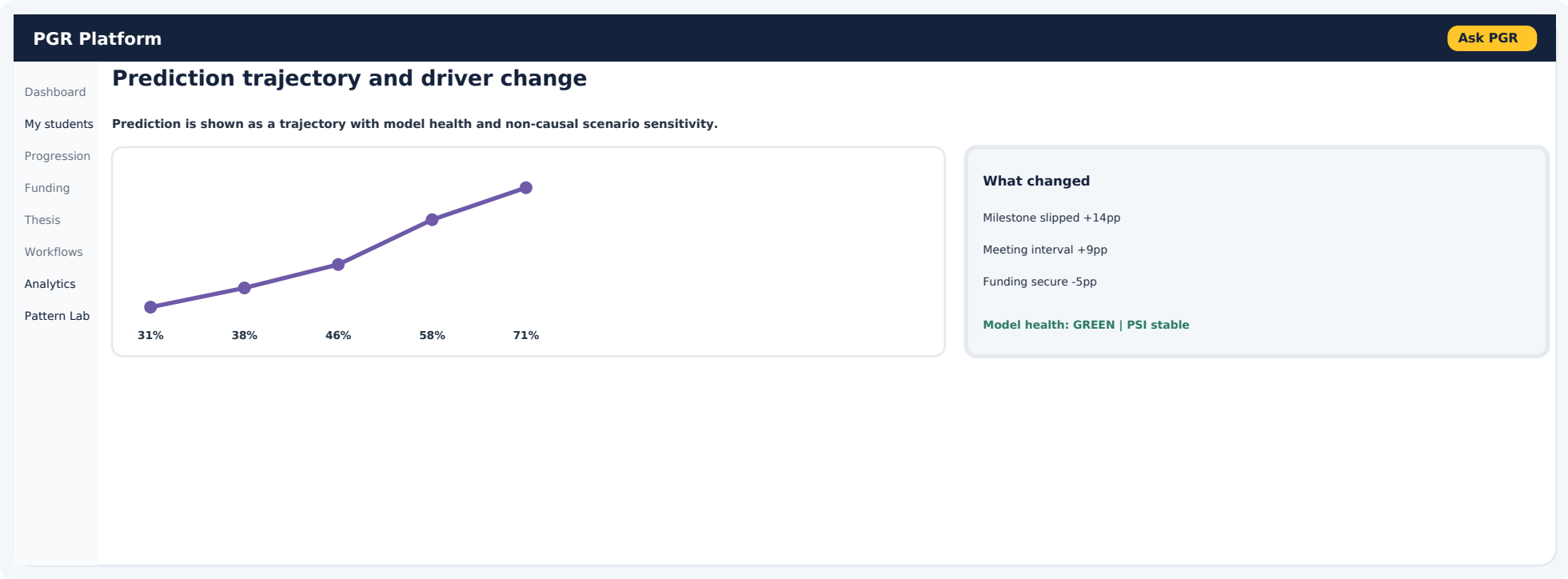
PGR Platform — User Manual Page 27



Current screen anchor	Upgrade direction
Current UI: Pattern Lab advanced screen from the supplied manual.	Target: keep the same navigation, route context and domain cards. Add intelligence in-line rather than creating a separate AI application.

13. Pattern Lab student panel - detailed layout

The first thing users should see is how the score moved, not just the current probability. Model health must remain visible enough that a degraded model cannot look equally trustworthy.



Region	Content
Trajectory chart	6-12 historical predictions, threshold line, model-version boundaries and current point.
What changed	Top driver deltas since prior snapshot; clearly labelled as model explanation.
Model health	Production model version, PSI, matured performance/calibration health and review state.
Explore scenarios	Bounded sensitivity controls; permanent non-causal banner.
Evidence / data	Source feature values and freshness; allow user to flag incorrect data rather than edit model output directly.

14. Ask PGR - current anchor -> Case Copilot

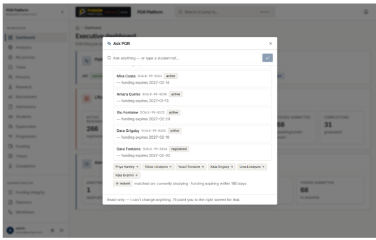
Keep the global palette and fast deterministic experience. In student/cohort context, Ask PGR should open in a case-aware mode with visible working context and a staged action plan, not a blank chatbot window.

Ask PGR — natural-language layer

What it solves. A rules-first assistant. Ask a question in your own words — who is falling through the cracks, whose money is running out, no supervision meeting in 90 days — and the platform understands, executes the equivalent filter, and shows the students. The LLM is off by default; the concept graph and typo-corrected parser cover most real questions with no student data leaving the server. Read only in this release.

How to use it.

- Launch it from the floating **Ask PGR** button or press **Ctrl/Cmd+K**.
- Type your question; the panel shows results as you type.
- Every answer carries an **Understood as** readback so you can verify the interpretation.
- Click a suggested student to open their record; the assistant does not do writes.



Ask PGR answering "funding expiring in 6 months" — matched deterministically in ~1 ms.

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Ask PGR - Case Copilot

How is Marcus doing, and what should we do next?

3 signals converged

review due in 18 days

82 days since meeting

drift score increased 17pp

CASE CONTEXT

Student: Marcus Bell

Window: current term

Pending: none

[Prepare 3 actions]

Current screen anchor	Upgrade direction
Current UI: Ask PGR natural-language layer from the supplied manual (manual shows earlier read-only release; current inventory describes confirmed writes).	Target: keep the same navigation, route context and domain cards. Add intelligence in-line rather than creating a separate AI application.

15. Ask PGR Case Copilot - conversation anatomy

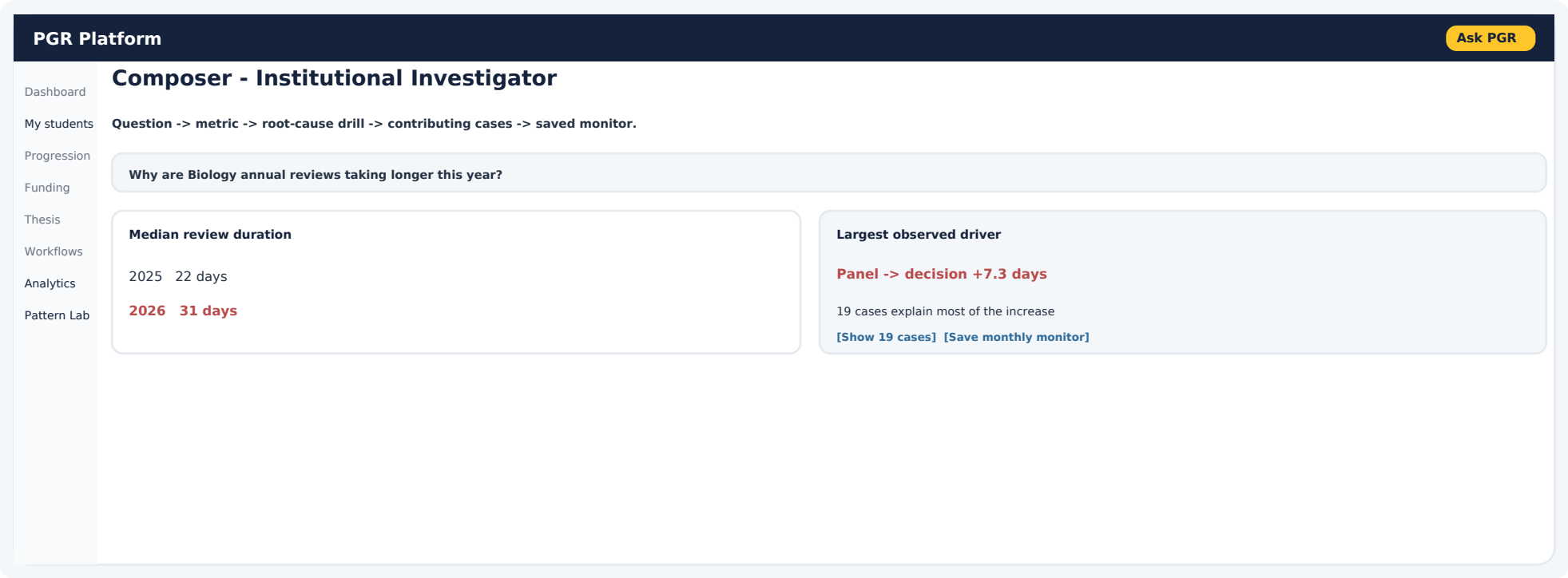
Case Copilot should show what it has resolved so users can correct context before an action plan is prepared.



UI element	Purpose
Case context rail	Resolved student/cohort, time window, filters and pending plan.
Answer body	Short conclusion first; evidence chips inline; no wall of prose.
Interpretation readback	Keep the existing "Understood as" concept for ambiguous requests.
Action proposal	Appears as a card with exact commands; opens confirmation drawer.
Trace / Why	Technical routing trace stays secondary; Evidence Drawer is the primary user explanation.
Model-off state	Same context rail and deterministic results; label AI narration unavailable.

16. Composer -> Institutional Investigator

Composer should feel like an analyst following a controlled drill path. The user asks once, then explores why, where and which cases without restarting from a blank prompt.



Interaction	Behaviour
Question	Natural language picks trusted catalogue functions.
Result blocks	Charts/tables render from deterministic function output.
Narrative	Explains the largest observed change with evidence chips.
Drill chips	"By department", "By stage", "Show 19 cases" are allow-listed next steps.
Case handoff	Opening a contributor switches to student Case Copilot context.
Monitor	Save function graph + parameters as scheduled dashboard/alert definition.

17. Research Change Radar

The comparison view should be visual and review-oriented. Show Earlier and Current passages side by side, then list material changes and possible implications as review flags.

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Research Change Radar - document comparison

Research Change Radar

Material change detected between Proposal v2 and Annual Review v1

Earlier

Qualitative interviews

40 participants

Objective 3 included

Current

Mixed methods

120 participants

Objective 3 unclear

Review implications

Component	Detailed UX
Document selector	Explicit version A / B; show upload dates and extraction status.
Change summary	Count by objective/method/population/scope/timeline; no "breach" label.
Side-by-side passages	Highlight exact supporting spans with page/section links.
Finding disposition	Accept finding / dismiss / add note; disposition persists.
Implication action	"Request ethics review" or "Discuss at progression review" only if supported as an action template, never automatic.
Model-off	Keep manual side-by-side compare and previously saved reviewed findings.

18. AI Evidence Drawer

Every AI card, narrative sentence, ranking and document finding should be able to open one consistent evidence drawer. This is more useful to end users than a raw prompt trace.

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Evidence Ledger - source-linked explanation

Every conclusion expands into a source-linked evidence ledger.

Claim	Source	Value / passage	Freshness	Status
Review due soon	Milestone M-1882	Due 29 Sep	today	verified
Meeting gap	Meeting S-991	Last 15 Jul	today	verified
Funding concern	Funding F-201	Ends 31 Jan 2027	today	verified

Research plan: Annual Review v1. Methods changed 24 March 2024. Review generated facts: 0 | Missing information: Finance inbound confirmation unavailable

Drawer section	Detail
Verified facts	Source label, current value, as-of time, open-record link.
Model signals	Model version, probability/threshold, health status, prediction timestamp.
Document findings	Version, page/section locator, matched passage and review status.
Missing information	Explicit source gap and which conclusion it limits.
Context overrides	Actor, timestamp, value and scope.
Technical trace	Collapsed advanced section: router tokens, app/ai call class, prompt template version, fallback path.

19. Institutional Memory - comparable case explorer

Use a case explorer, not a "recommend outcome" button. Users should understand why cases are considered comparable and what evidence was commonly requested.

Panel	UI content
Similarity summary	Matched on request type, study mode, funding situation, timing and milestone context.
Historical handling	Distribution of outcomes with date/policy window; avoid persuasive colour coding.
Common evidence	Most frequently requested documents/statements.
Case cards	Anonymised by default where possible; open only if current user is authorised.
Primary CTA	Use as context / open case - never "copy outcome".
Bias warning	Historical frequency is not a rule and may reflect old policy/practice.

20. Policy Compiler workspace

This should look like governed configuration management, not a chat prompt. Users need to compare the approved source, proposed system changes and affected cohort at the same time.

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Policy Compiler - source, mapping and impact

Three-column workspace: approved source -> proposed mapping -> affected population.

SOURCE POLICY

2027 PGR Progression

Regulations

Clause 4.2:

Annual review within 10 months

PROPOSED CHANGE

annual_review_month

12 -> 10

appeal_window_days

28 -> 21

IMPACT PREVIEW

143 students affected

37 milestones earlier

14 workflows overdue

0 decided milestones move

Review proposed change

Interaction	Detail
Source policy	Approved version metadata + clause viewer.
Proposed change	Structured old -> new config/rule card with confidence replaced by evidence/needs-review state.
Impact preview	Affected record counts, workflow/timeline effects and examples.
Approval	Per-change checkbox + reviewer note; second approval if institutional governance requires it.
Publish	Calls existing configuration mechanism; creates audit entry and rollback reference.
Unsupported clause	Explicit "Manual implementation required" card.

21. Proactive intelligence centre

Do not flood the user with AI notifications. A small Intelligence inbox or dashboard section should surface only material change, systemic bottlenecks and governance issues, each with a clear owner and route.

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PGR Intelligence Alerts

Proactive intelligence appears only for material changes, with suppression and ownership.

SYSTEMIC BOTTLENECK

14 annual reviews within 30d have no panel confirmed

Increased from 3 last week

Investigate

FUNDING COVERAGE

8 students now finish after recorded funding end

5 appeared after lifecycle changes

Review cases

MODEL HEALTH

Milestone drift AUC fell below review threshold

Production model requires governance review

Open Pattern Lab

Add suppression controls: do not resend unchanged alerts, group identical cases, respect suspended students and existing task/escalation state, and show why the alert was generated.

22. Loading, model-off, missing-data and failure states

Exceptional AI UX is defined by failure behaviour. These states should be designed before implementation, not added after the happy path.

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AI component states

Every AI component needs designed loading, model-off, missing-data and failure states.

LOADING

Facts render immediately. Narrative skeleton loads separately, the record.

MODEL OFF

Show deterministic result and "AI narration unavailable. Workflow remains usable.

MISSING DATA

Show what is missing and which conclusion is linked. Never fill gaps with model guesses.

GROQ ERROR

Keep evidence shell; allow retry or manual path. No duplicate actions.

STALE MODEL

Show model health banner and suppress overconfidence. Route to Pattern Lab review.

State	Copy rule
Model off	"AI narration is unavailable. Showing rule-based results."
Missing data	"Finance confirmation is not available, so payment receipt cannot be confirmed."
Stale evidence	"This explanation was generated before the milestone changed. Refresh analysis."
Stale model	"Prediction model is under review. Treat this score as advisory."
Job queued	"Document comparison queued. You can continue working; results will appear here."
Action conflict	"The record changed after this plan was prepared. Review the updated state before confirming."

23. Role-based visibility

The same component should render differently by role rather than creating separate AI products. Frontend visibility should mirror backend authorisation and never rely on hiding controls alone.

Role	Visible intelligence	Restricted / adapted
Institution admin	Full authorised student/cohort intelligence, policy and operational investigations	Still requires confirmation/approval for consequential actions
Supervisor	Own supervisees, meeting copilot, engagement, relevant progression/funding context	No unrelated student similarity/case data
Student	Own next steps, transparent deadlines/evidence, optional private practice features	No staff-only risk ranking/intervention metadata unless policy explicitly allows
Executive	Aggregated investigations and systemic alerts	Individual detail only where role permits
Registry/governance	Policy Compiler, Evidence Ledger, Pattern Lab governance	No bypass of domain approval separation

24. Frontend component map - Next.js / shadcn implementation

Keep the components composable and route-aware. The same EvidenceDrawer and ActionPlanDrawer should be used by Ask PGR, Weekly Review, student record and Composer.

Component	Purpose	Likely composition
IntelligenceStrip	Compact current state on student page	Card + badges + 4 action buttons
EvidenceDrawer	Uniform source/evidence view	Sheet/Drawer + table + source links
ScenarioDrawer	Current vs proposed rule/model results	Sheet + diff table + warning banner
ActionPlanDrawer	Review multi-action bundle	Sheet + checkbox rows + confirmation footer
TrajectoryChart	Engagement / prediction history	Accessible chart + tabular fallback
ChangeDiff	Earlier/current document comparison	Resizable panels + highlighted passages
CaseContextRail	Ask PGR working context	Compact side rail / chips
AIStateBanner	Model-off/stale/missing/error state	Alert component with clear recovery action
SourceChip	Inline evidence link	Badge/Button opening EvidenceDrawer

25. Data fetching and UI state

Frontend rendering should separate deterministic data from slower narrative. This keeps the app responsive and honours the existing model-off guarantee.

Layer	Recommended behaviour
Page load	Fetch student/domain/intelligence factual shell first. Render without waiting on Groq.
Narrative	Lazy-load/stream explanation into reserved region; do not shift critical controls.
Evidence	Fetch on drawer open or include lightweight refs in initial payload.
Scenario	Explicit POST from user action; cache only by exact input hash and source state version.
Intervention plan	Draft is server-generated; local UI can edit allowed fields before confirm.
Confirm	Disable duplicate submit; show optimistic progress only after server accepts idempotency key.
Background jobs	Poll/SSE/websocket according to existing app conventions; job card survives navigation.

26. Accessibility and responsive behaviour

The intelligence layer must remain understandable without colour, hover or desktop-only layouts.

Area	Requirement
Keyboard	All Explain/Evidence/Simulate/Prepare actions reachable; drawers trap/focus correctly.
Colour	Status includes text/icon, not colour alone.
Charts	Provide tabular values and accessible summaries for trajectories.
Evidence	Source links have descriptive labels and visible focus.
Mobile	Intelligence strip collapses to current state + one primary action; drawers become full-screen sheets.
Tablet	Single-column cards; sticky action bar for Evidence/Simulate/Prepare.
Motion	Respect reduced-motion preferences; avoid animated "AI thinking" gimmicks.

27. Frontend delivery sequence

Build one shared interaction grammar before adding every feature. The UI becomes easier to learn and the engineering surface stays manageable.

Sprint / release focus	Frontend deliverables
Foundation	IntelligenceStrip, EvidenceDrawer, SourceChip, AIStateBanner, CaseContextRail
Existing feature upgrades	Case Copilot panel, Weekly Intervention Queue, Meeting commitment review, Engagement Trajectory
Simulation	ScenarioDrawer + Twin timeline + rule/model separation patterns
Pattern Lab	Prediction trajectory, model health banner, sensitivity explorer
Documents	ChangeDiff, document locator viewer, finding disposition controls
Institutional tools	Comparable case explorer, Policy Compiler three-column workspace, proactive Intelligence inbox

28. Frontend source alignment and assumptions

This UX specification is aligned to the supplied running-app screenshots and architecture: current navigation, student record, Pattern Lab, Ask PGR, Next.js 14, Tailwind/shadcn Redwood Professional, row scoping and existing domain routes are preserved. Proposed screens/components are upgrades, not claims about current implementation.

- Manual page 10 is used as the current student-record anchor; page 27 as Pattern Lab; page 28 as the earlier Ask PGR visual reference.
- The user-provided current AI inventory supersedes the manual where it says Ask PGR is read-only: current Ask PGR is described as supporting confirmed actions.
- The current inventory describes /reviews/weekly, /my-students, /composer and /assistant; the wireframes keep those surfaces conceptually rather than inventing a new navigation tree.
- Document semantic comparison, Institutional Memory and Policy Compiler need backend foundations described in PDF 2; the frontend should not fake these experiences before evidence/permission services exist.
- PGR_Platform_Critical_Gaps.pdf was referenced by the supplied platform documents but was not provided; any frontend implications from those gaps remain outside this specification.